Physics problem solving

Below are some general strategies to solving physic problems:

- **Draw a diagram (if appropriate).**
  - **Clearly Label all relevant quantities**
  - This can be absolutely critical for certain types of problems (e.g., free-body diagrams)
  - physics is physical, having a picture of the system can transform a challenging problem into an easy one.
- **Write down what you know, and what you are trying to find out.**
  - For example, if there are three unknowns that you're trying to find, but you've written down only two facts, then you know there must be another fact you're missing (like a conservation law, or an $F = ma$ equation)
- **Solve things symbolically.**
  - If you are solving a problem where the given quantities are specified numerically, start by changing the numbers to letters and solve the problem in terms of letters. After you obtain an answer in terms of letters, you can plug in the actual number values to obtain a numerical answer.
  - Advantages to using symbols:
    - It's quicker; it's easier to multiply letters, than to multiple numbers
    - You're less likely to make mistakes.
    - You can check units and special cases
- **Pay attention to units and make sure they are correct.**
  - If you are asked for a velocity but your formula gives you something else, you know you made a mistake somewhere.
- **Check order of magnitude if you end up with a numerical answer.**
  - If you end up with a actual numerical answer, do a sanity check to see if the resulting number is reasonable.
  - For example, if you calculated the distance a car takes to come to a stop and you get an answer in km or mm then you probably did something wrong.